

Compost Pile Analyzer

Quickstart & Engineering Reference Guide



1. What Is Composting?

Composting is the **managed decomposition** of organic residues into more stable organic compounds — a natural biological process accelerated and controlled by the engineer. Under ideal conditions it is an **aerobic process**, meaning oxygen-breathing microorganisms drive the transformation. The finished product is biologically and chemically stable, **homogeneous**, with raw feedstocks no longer distinguishable.

Who does the work?

Composting is a two-stage biological effort. Larger **soil fauna** — earthworms, nematodes, mites, sowbugs, beetles, springtails, and ants — are responsible for much of the initial *mechanical breakdown* of materials into smaller particles. Once particle size is reduced and surface area increased, **bacteria, fungi, actinomycetes, and protozoa** colonize the organic material, beginning the actual chemical composting process.



2. Composting Phases



Pile formation → Stable humic compost

Active Phase	Curing Phase
Temperature: Rises rapidly to 130–150°F — high enough to kill most pathogens including <i>E. coli</i> , <i>S. aureus</i> , <i>B. subtilis</i> , and <i>C. botulinum</i> .	Temperature: Declines to ~100°F as readily-available carbon is exhausted.
Duration: Begins 24–72 hours after pile formation; lasts 1–4 weeks depending on management and feedstock.	Duration: 1–4 months. Finished compost is biologically and chemically stable.
Oxygen: Must be replenished continuously via passive aeration, forced aeration (blowers), or manual pile turning.	Oxygen: Consumption declines — pile can be stockpiled without turning. Organic materials continue decomposing into stable humic substances .

3. Optimal Conditions

Parameter	Optimal Range
C:N Ratio	25:1 – 35:1
Moisture	45–60% by weight
Oxygen	>10% available O2
Bulk Density	~1,000 lbs/cu yd
pH	6.5 – 8.0
Active Temp.	130 – 150°F
Curing Temp.	~100°F

4. Tool User Guide

The Compost Pile Analyzer is a browser-based tool that calculates the blended carbon-to-nitrogen (C:N) ratio and moisture content of a multi-material compost pile and evaluates whether conditions are within optimal ranges. Each organic waste category is assigned a fixed C:N ratio and moisture content based on typical literature values (shown in the Notes column of the inputs table); actual waste characteristics may vary depending on source, season, and handling. If conditions are not within optimal ranges, the tool recommends material additions to correct the imbalance. The guide below explains inputs, calculations, outputs, and how to act on results.

4a. Inputs

Input	Unit	Notes
Kitchen scraps mass (M_k)	lbs	C:N 15:1; 70% moisture
Yard waste mass (M_y)	lbs	C:N 40:1; 60% moisture
Paper/cardboard mass (M_p)	lbs	C:N 150:1; 10% moisture
Raw animal manure mass (M_m)	lbs	C:N 15:1; 75% moisture; triggers regulatory flag if >0 lbs
Pile temperature	°F	Enter current measured pile temperature
Composting phase	Active / Curing	Select based on pile age and temperature trend

4b. Engineering Calculations

All calculations follow a mass-weighted blending approach. Each material is first decomposed into its carbon (C) and nitrogen (N) mass fractions using its known C:N ratio R, then totaled across all materials.

Step 1 — Carbon and Nitrogen fractions for a material with ratio R:

$$\text{Carbon fraction} = R / (R + 1) \quad \text{Nitrogen fraction} = 1 / (R + 1)$$

Step 2 — Total Carbon across all materials:

$$C_{\text{total}} = M_k \times (15/16) + M_y \times (40/41) + M_p \times (150/151) + M_m \times (15/16)$$

Step 3 — Total Nitrogen across all materials:

$$N_{\text{total}} = M_k \times (1/16) + M_y \times (1/41) + M_p \times (1/151) + M_m \times (1/16)$$

Step 4 — Blended C:N ratio:

$$\text{C:N} = C_{\text{total}} / N_{\text{total}} \text{ (rounded to nearest whole number)}$$

Step 5 — Weighted average moisture:

$$\text{Moisture (\%)} = [0.70 \cdot M_k + 0.60 \cdot M_y + 0.10 \cdot M_p + 0.75 \cdot M_m] / (M_k + M_y + M_p + M_m) \times 100$$

Adjustment Equations

If the pile is out of optimal range, the tool solves for the mass x of a corrective material needed to reach a target C:N of 30:1 (midpoint of the optimal 25–35 range).

Case 1 — Too nitrogen-rich (C:N < 25): Add paper/cardboard

$$x = (30 \cdot N_{\text{total}} - C_{\text{total}}) / (120/151)$$

Derived from solving: $(C_{\text{total}} + x \cdot 150/151) / (N_{\text{total}} + x \cdot 1/151) = 30$

Case 2 — Too carbon-rich (C:N > 35): Add kitchen scraps

$$x = (C_{\text{total}} - 30 \cdot N_{\text{total}}) / (15/16)$$

Derived from solving: $(C_{\text{total}} + x \cdot 15/16) / (N_{\text{total}} + x \cdot 1/16) = 30$

Moisture adjustments

If too wet (>60%): Add paper/cardboard (10% moisture) in increments — the moisture and C:N interact, so the tool provides guidance to add gradually and re-measure.

If too dry (<45%): Add kitchen scraps (70%) or yard waste (60%) and re-measure.

4c. Outputs

After clicking *Analyze compost pile* the tool produces three result panels:

- **Calculated Outputs** — The blended C:N ratio (rounded whole number, e.g. 28:1) and moisture percentage (one decimal place, e.g. 54.3%) displayed as large metric cards.
- **Feasibility Assessment** — Each parameter (C:N, moisture, temperature) is evaluated against optimal ranges and labeled *Optimal*, *Not hot enough*, *Too hot*, *Not stabilized yet*, or the specific out-of-range condition (e.g. Too nitrogen-rich, Too carbon-rich, Too wet, Too dry). Color-coded badges (green / yellow / red) allow rapid interpretation.
- **Recommended Adjustments** — Shown only when C:N or moisture is out of range. Provides a calculated mass (lbs) of material to add, or practical guidance for moisture correction.
- **Regulatory Flag** — Appears automatically if raw manure mass > 0 lbs, reminding the user of the 90–120 day pre-harvest interval requirement.

4d. Interpreting Outputs

Result	What it means	Recommended action
C:N too low (<25:1)	Pile is nitrogen-rich; likely to produce ammonia odors and lose N	Add paper/cardboard per tool recommendation
C:N too high (>35:1)	Pile is carbon-rich; decomposition will be slow	Add kitchen scraps per tool recommendation
Moisture too wet (>60%)	Poor aeration; anaerobic conditions may develop	Add dry paper/cardboard; improve drainage
Moisture too dry (<45%)	Microbial activity inhibited	Add kitchen scraps or yard waste; consider irrigation
Temp. too low — active (< 130°F)	Decomposition process slows or stops	Increase aeration (turn pile); check C:N and moisture
Temp. too high — active (>150°F)	Microbial inhibition risk	Turn pile to release heat; increase aeration
Not stabilized yet — curing (>100°F)	Pile may still be in active phase	Allow more time or reduce pile size

■ Caveats and Limitations

- C:N ratios and moisture values are literature-based estimates. Actual feedstock values vary by source, season, and handling.
- The tool assumes uniform mixing. Layered piles will have spatial variation not captured by these blended calculations.
- Temperature inputs are user-reported. Accuracy depends on thermometer placement and calibration.
- Moisture adjustment is iterative — the tool provides guidance rather than a precise closed-form answer because adding one material changes both C:N and moisture simultaneously.
- Raw manure regulatory requirements vary by jurisdiction. Consult local regulations and the applicable state department of agriculture.